

# Sahil Singh

RESEARCH / SCIENTIFIC ML / EMBEDDED SYSTEMS

## PROFILE

Student researcher working across physics-informed machine learning, biomedical inverse modeling, world-model evaluation, and human-validated satellite imagery.

## 01 Experience

### Research Assistant | MIT STAR Lab

SEP 2026 - PRESENT

Validate news-driven satellite-event detections using temporal imagery; adjudicate visibility windows and document uncertain cases for SkyScraper's human-in-the-loop geospatial pipeline under Prof. Kerri Cahoy and Ms. Anderson.

### Research Assistant | Rochester Institute of Technology

JUL 2026 - PRESENT

Study ventricular tachycardia in Prof. Linwei Wang's lab under Sumeet Vadhavkar. Built a reproducible openCARP S1-S2 scar-substrate study, resolving the capture transition at 368-369 ms and a 46 ms directional delay across 24/24 ring sectors.

## 02 Accepted Research

### GEML

NEURIPS VERICODEGEN

Studied learned equivalence over expressions compiled to one operator; measured 41x median tree expansion and showed that nine fitted graph models remained near chance under variable-role swaps.

### MineOcclude

NEURIPS PHYSWORLDAI + ESR

Built a paired Minecraft benchmark separating visual evidence, motion predictability, and readout effects; achieved 0.083-0.107 block RMSE under visible/glass conditions and characterized failure under opacity.

### MIG-PINO Cardiac Field Localization

NEURIPS PHYSWORLDAI

Evaluated FNO and DeepONet on 500 cardiac simulations: held-out LAT/APD90 field R2 reached at least 0.954 while localization Dice exposed failure modes hidden by global accuracy.

## 03 Selected Engineering & Leadership

- CubeSAT Power Subsystem Co-Lead - modeled generation, storage, duty cycles, and loads; presented at SmallSat 2026 and the NCSS Student Research Conference 2026.
- CASCADE, MIT Blueprint - combined an RNN trained on 7,629 outbreak records with a global SEIR simulator and Monte Carlo Tree Search across 155+ countries.
- Distributed gas-leak localization - ESP32 PM2.5/CO2 sensor graph, dual Raspberry Pi 5 cluster, and lightweight U-Net spatial reconstruction.
- AP Calculus BC teaching assistant and Math Olympiad tutor; Leading Officer at Peachtree Math Circle.

## 04 Additional Projects

- Passive edge-IoT energy harvester - piezoelectric and triboelectric multi-source design; GSEF Best in Category and creative problem-solving award.
- Autonomous agriculture drone - custom 3.5-inch build with ArduPilot flight control, Arduino seed payload, and soil-informed deployment; CPS Expo 2nd place.
- IRIS face-recognition attendance system - Raspberry Pi inference with automated spreadsheet logging.

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DULUTH, GA  
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## CONTACT

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### Portfolio

gensahilsingh.vercel.app

### LinkedIn

linkedin.com/in/sahil-singh-17a641239

### GitHub

github.com/sahilsinghthefirst

## EDUCATION

### Georgia Institute of Technology

Distance Math Year 1 + CS 1301  
Aug 2026 - Present

### Fulton Science Academy

Grade 10 | Expected May 2029

## TECHNICAL SKILLS

### Machine Learning

PyTorch, neural operators, FNO, DeepONet, GNNs, U-Net, RNNs, Monte Carlo dropout

### Scientific Computing

openCARP, CellML, BOCF, Biot-Savart modeling, inverse problems, uncertainty quantification

### Programming

Python, C++, Java, Linux/WSL, reproducible research workflows

### Embedded & Robotics

Raspberry Pi, ESP32, Arduino, sensors, soldering, drones, ArduPilot, CAD/Blender

## RECOGNITION

- 2x GSEF Best in Category
- Thermo Fisher JIC Top 300
- 3x Fulton County Science Fair qualifier
- GaSTC state qualifier, grades 6-8; 1st in category twice
- Top 3 Speaker, Ivy Bridge Debate